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1  # Note 03: Finding the Best Fitted Line (R_Squared Method / Adjusted R_Squared Value)
2  # =====
3
4  # Import Libraries
5  import numpy as np
6  import matplotlib.pyplot as plt
7  import pandas as pd
8
9  # Load Dataset
10 df = pd.read_csv("test_data.csv")
11
12 # Define X and Y Variables
13 x = df['temperature']
14 y = df['volt']
15
16 # Define Empty Lists To Store R2 Values and Define the Order
17 R2 = []
18 orders = range(1, 6)
19
20 # Draw Elbow Plot
21 for order in orders:
22     # Fit polynomial
23     coefficient = np.polyfit(x, y, order)
24     # Create polynomial function
25     f = np.poly1d(coefficient)
26     # Predicted y values
27     yp = f(x)
28     # R2 calculation
29     ss_res = np.sum((y - yp) ** 2)
30     ss_tot = np.sum((y - np.mean(y)) ** 2)
31     # Finding R_Squared
32     r_squared = 1 - (ss_res / ss_tot)
33     # Append R_Squared Value
34     R2.append(r_squared)
35
36 # Plot R2 vs Order
37 plt.figure()
38 plt.scatter(orders, R2, label="R2", color="r", marker="o")
39 plt.plot(orders, R2, label="R2", color="b")
40 plt.xlabel("Order")
41 plt.ylabel("R2")
42 plt.title("R2 vs Polynomial Order")
43 plt.grid(True, alpha=0.3)
44 plt.show()
45
46 # Print Best Order
47 print("Best Order : 3")
48
49 # Plot the Smooth Curve
50 plt.figure()
51 plt.scatter(x, y, label="Original Data Points", marker="o", color="r")
52 x_new = np.linspace(min(x), max(x), 1000)
53 coefficient = np.polyfit(x, y, 3)
54 f = np.poly1d(coefficient)
55 yp = f(x_new)
56 plt.plot(x_new, yp, label="Best Fitted Line", color="g")
57 plt.xlabel("x")
58 plt.ylabel("y")
59 plt.title("Best Fitted Polynomial")
60 plt.grid(True, alpha=0.3)
61 plt.legend()
62 plt.show()
63
64 # Find Adjusted R_Squared Values
65 R_Squared_Value = R2[2]
66 n = 31 # Number of Observations
67 p = 1 # Number of Independent Variables
68 Adjusted_R_Squared_Value = (1 - ((1 - R_Squared_Value) * ((n - 1) / (n - p - 1))))
69 print("Adjusted R_Squared_Value:", Adjusted_R_Squared_Value)

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